Xanthogranulomatous Pyelonephritis (XGP): Comprehensive Disease Characterization

MONDO ID: MONDO:0007022 · **Category:** Infectious / Chronic Inflammatory Renal Disease **ICD-10:** N11.8 / N10 (chronic pyelonephritis) · **MeSH:** Pyelonephritis, Xanthogranulomatous (D058365) **Evidence base:** 11 confirmed findings across 42 reviewed papers (multicenter series, pooled analyses, case series, and case reports)

Summary

Xanthogranulomatous pyelonephritis (XGP) is a rare, acquired (non-genetic) chronic destructive granulomatous form of pyelonephritis. It arises from a characteristic triad—**chronic urinary tract obstruction** (usually by staghorn/struvite calculi), **recurrent bacterial infection** (chiefly *Escherichia coli* and urease-producing *Proteus mirabilis*), and an **incomplete/aberrant host immune response**—that together drive the progressive replacement of renal parenchyma by **lipid-laden CD68+ foamy macrophages**, granulomatous inflammation, and fibrosis. The end result is an enlarged, usually unilateral, non-functioning kidney. The disease predominantly affects middle-aged women and spans all ages from neonate to elderly.

XGP is clinically and radiologically a "great imitator," most notably mimicking renal cell carcinoma (RCC). The classic CT sign is the **"bear paw"** appearance with a central staghorn calculus. Because imaging and even gross pathology overlap with malignancy, **definitive diagnosis is histopathologic**, and **immunohistochemistry (CD68)** is frequently decisive in distinguishing XGP foam cells from the clear cells of RCC. Patients typically present with flank pain, fever, and recurrent urinary tract infections, accompanied by anemia, leukocytosis, pyuria, and hematuria, often after a long diagnostic delay (median ~1 year in one series).

Management is stratified by disease extent and patient age: **nephrectomy remains the definitive treatment for adult diffuse XGP**, whereas **conservative antibiotic-based (kidney-sparing) management** succeeds in a majority of pediatric and focal cases. The

condition is curable but carries substantial perioperative morbidity (postoperative complication rates up to 73%, with half being Clavien-Dindo grade ≥3 in one series) and risks of local invasion—psoas abscess, perinephric abscess, and fistulae to skin, colon, or even lung. Rare coexistence with renal malignancy (squamous cell carcinoma, urothelial carcinoma, RCC) is documented, driven by the shared background of chronic stone disease and inflammation. XGP has also now been documented as a naturally occurring disease in at least one non-human species (dog).

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1. Disease Information

Overview. XGP is an uncommon chronic destructive granulomatous process of the renal parenchyma occurring in association with long-term urinary tract obstruction and infection. It affects females more often than males across a wide age range, from newborn to elderly PMID: 21526966. It is regarded as a rare variant of chronic pyelonephritis that produces an enlarged, non-functioning kidney with diffuse parenchymal damage.

Key identifiers. - **MONDO:** MONDO:0007022 - **MeSH:** Pyelonephritis, Xanthogranulomatous - **ICD-10:** grouped under chronic pyelonephritis (N11.x) - **OMIM / Orphanet:** No Mendelian OMIM entry (XGP is acquired, not a monogenic disease). No distinct genetic locus.

Synonyms / alternative names: Xanthogranulomatous pyelonephritis (XGP, XGPN, XPN); xanthogranulomatous pyelonephritis of the kidney. In older literature sometimes described under "foam-cell" or "lipogranulomatous" pyelonephritis.

Source of information. Evidence is derived predominantly from **aggregated disease-level resources**: retrospective multicenter series (e.g., 365 patients across 10 centers/5 countries; PMID: 37242365), pooled quantitative analyses of published cases (251 patients; PMID: 39254583), single-institution case series, and individual case reports. There is no dedicated genetic or population registry; there are no EHR-derived individual-patient omics datasets specific to XGP.

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2. Etiology

Primary causal factors. XGP is a fundamentally **acquired, infectious/mechanistic** disease. It is thought to arise secondary to a combination of **obstruction, recurrent bacterial infection, and an incomplete immune response**, resulting in focal or diffuse renal destruction PMID: 34522121. Ultrastructural evidence supports a **bacteria-induced process** whose histologic appearance relates to incomplete bacterial degradation and altered host response PMID: 7076194.

Risk factors (environmental / clinical). - **Nephrolithiasis / staghorn calculi** — the dominant risk factor; multiple stones present in 34.5% of a 365-patient series PMID: 37242365, and renal stones in 53.8% of a 251-patient pooled cohort PMID: 39254583. - **Chronic urinary obstruction** (stones, stricture, rarely tumor). - **Recurrent urinary tract infection.** - **Female sex** — 57–72% of patients across series PMID: 37242365; PMID: 39254583; PMID: 36505992. - **Chronic kidney disease** — most common comorbidity (71%) PMID: 37242365. - **Diabetes mellitus / immunocompromise, low socioeconomic status, and limited healthcare access** are reported associations in case literature.

Genetic risk factors. None established. XGP is not a heritable Mendelian disorder; there is no OMIM entry, no ClinVar variants, and no GWAS signal specific to XGP. Rare occurrence in patients with polycystic kidney disease is reported but reflects anatomic predisposition to obstruction/infection rather than a defined genetic driver PMID: 39185472.

Protective factors. No genetic protective variants are known. Environmentally, prevention of the causal chain—effective treatment of urolithiasis, relief of obstruction, and prompt treatment of UTIs—is protective by preventing the disease from developing.

Gene–environment interactions. Not applicable in a defined molecular sense. The functionally relevant "interaction" is between **host immune competence** (an incomplete/aberrant macrophage response) and the **environmental insult** (obstruction + chronic infection), which together determine whether chronic infection progresses to xanthogranulomatous destruction PMID: 7076194; PMID: 34522121.

3. Phenotypes

XGP presents with a stereotyped constellation of symptoms, signs, and laboratory abnormalities. The best-quantified profiles come from a 57-patient case series PMID: 36505992 and a 55-case series PMID: 41250779.

Phenotype	Type	Frequency	Suggested HPO term
Flank / lumbar pain	Symptom	89.5%	Flank pain (HP:0030157)
Fever	Symptom/sign	43.9–55.0%	Fever (HP:0001945)
Recurrent UTI	Clinical course	43.9%	Recurrent urinary tract infections (HP:0000010)
Anemia	Lab abnormality	78.9%	Anemia (HP:0001903)
Leukocytosis (± left shift)	Lab abnormality	43.6% (left shift 21.6%)	Leukocytosis (HP:0001974)
Hematuria	Lab abnormality	75.6%	Hematuria (HP:0000790)
Pyuria / leukocyturia	Lab abnormality	93.2%	Pyuria (HP:0012085)
Bacteriuria	Lab abnormality	40.9%	Bacteriuria (HP:0025427)
Palpable abdominal / flank mass	Physical sign	variable	Abdominal mass (HP:0031183)
Renal calculi	Structural	~54% to 49/55 (~89%)	Nephrolithiasis (HP:0000787)
Hydronephrosis	Structural	common	Hydronephrosis (HP:0000126)
Weight loss	Constitutional	variable	Weight loss (HP:0001824)

Direct evidence. "The most common symptoms were flank/lumbar pain (89.5%), fever (43.9%), and recurrent urinary tract infection (43.9%)" and "Blood tests showed anemia (78.9%), leukocytosis (43.6%) with left shift (21.6%). Urinalysis showed hematuria (75.6%), bacteriuria (40.9%), and leukocytes (93.2%)" PMID: 36505992. Reviews additionally list lower urinary tract symptoms, palpable mass, gross hematuria, and weight loss PMID: 21526966.

Phenotype characteristics. - **Age of onset:** predominantly adult (mean age 36–49 years across series), but the full range spans neonate to elderly PMID: 21526966. Pediatric disease occurs and is often focal. - **Severity:** moderate to severe; can be indolent/asymptomatic until late-stage, then fulminant. - **Progression:** chronic and progressive with parenchymal destruction; frequently insidious with a long diagnostic delay (median ~365 days in one series) PMID: 36505992.

Quality-of-life impact. No disease-specific EQ-5D/SF-36 data exist for XGP. Impact is inferred from chronic pain, recurrent infection, anemia/fatigue, prolonged diagnostic delay, and the morbidity of eventual nephrectomy (loss of a kidney, perioperative complications).

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4. Genetic / Molecular Information

Causal genes: None. XGP is an acquired infectious/inflammatory disease with **no identified causal gene, pathogenic variant, or chromosomal abnormality**. There are no relevant OMIM, ClinVar, HGMD, or COSMIC entries specific to XGP.

Pathogenic variants / allele frequencies / somatic-vs-germline: Not applicable.

Modifier genes: None established. Host factors modulating macrophage lipid handling and bacterial clearance are mechanistically implicated but not genetically defined PMID: 7076194.

Epigenetic information: No XGP-specific DNA methylation, histone modification, or chromatin studies are available.

Chromosomal abnormalities: Not applicable to XGP itself. (Note: coexisting malignancies such as squamous cell or urothelial carcinoma have their own genomic alterations, but these are separate neoplastic entities.)

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5. Environmental Information

Environmental / mechanical factors. The dominant non-genetic contributors are **obstructive nephrolithiasis** (especially staghorn/struvite calculi) and **chronic urinary obstruction**, which create stasis and a nidus for persistent infection.

Lifestyle / clinical factors. Recurrent/neglected UTIs, low fluid intake, infrequent voiding, diabetes, and limited healthcare access (delaying stone/UTI treatment) contribute in reported cases.

Infectious agents. XGP is driven by chronic bacterial super-infection. The predominant organisms are ***Escherichia coli*** and ***Proteus mirabilis***, with occasional *Pseudomonas* and *Providencia stuartii*.

Organism	Evidence	NCBI Taxon
Escherichia coli	14.8% of a 27-case series; leading organism in reviews PMID: 33827527; PMID: 21526966	562
Proteus mirabilis (urease-producing)	18.5% of the same series PMID: 33827527; PMID: 34715795	584
Pseudomonas spp.	occasional PMID: 34715795	286
Providencia stuartii	rare, case report PMID: 34715795	588

Direct evidence. "Urine cultures most often reveal Escherichia coli and Proteus mirabilis" PMID: 21526966; and XGP is "most commonly associated with super-infections by bacteria such as E. coli, Proteus mirabilis, and occasionally Pseudomonas species" PMID: 34715795. Importantly, **kidney (upper-tract) urine culture is far more often positive (81.9%) than bladder urine culture (53.2%)** in the same patients, underscoring that the infection is compartmentalized in the obstructed kidney PMID: 37242365.

The urease activity of *Proteus mirabilis* is mechanistically relevant: it alkalinizes urine and promotes struvite (magnesium ammonium phosphate) stone formation, reinforcing the obstruction–infection loop.

6. Mechanism / Pathophysiology

Ordered causal chain

1. **Chronic urinary tract obstruction** (usually a staghorn/struvite calculus, less often stricture or tumor) → **leads to** urinary stasis and increased intrarenal pressure.
2. Stasis → **promotes** persistent bacterial colonization/infection, chiefly by *E. coli* and urease-producing *Proteus mirabilis* (which further drives struvite stone growth, reinforcing obstruction — a feed-forward branch).
3. Chronic infection → **triggers** a suppurative then granulomatous inflammatory response with recruitment of neutrophils, lymphocytes, plasma cells, and macrophages.
4. **Incomplete bacterial degradation and an altered/aberrant host immune response** → **results in** persistence of bacterial and lipid debris within macrophages (*inferred from ultrastructural data*: bacteria seen within PMNs and vacuoles, histiocytes laden with lipid droplets and phagolysosomes) PMID: 7076194.
5. Defective macrophage lipid clearance → **leads to** accumulation of lipid within histiocytes, forming characteristic **CD68+ lipid-laden "foam cells" (xanthoma cells)** PMID: 7076194; PMID: 42288880.
6. Progressive foam-cell accumulation with granulomatous inflammation and multinucleated giant cells → **causes** replacement and destruction of renal parenchyma, with fibrosis PMID: 42110084; PMID: 21526966.
7. Parenchymal destruction → **produces** an enlarged, non-functioning kidney (clinical XGP).
8. **Branch (local invasion):** transmural inflammation → **extends** into perinephric fat and adjacent structures → psoas abscess, perinephric abscess, and fistulae (nephrocutaneous, renocolic, and rarely transdiaphragmatic to lung) PMID: 36105829; PMID: 29480882.

Ultrastructural / histologic architecture

An ultrastructural study of 5 cases demonstrated **stratified xanthogranulomas**: a central purulent zone containing bacteria within PMNs and vacuoles, surrounded by a layer of lipid-droplet-laden histiocytes (foam cells) with phagolysosomes, then macrophages, lymphocytes, plasma cells, and peripheral fibroblasts PMID: 7076194. "Our study suggests that xanthogranulomatous pyelonephritis is a bacteria induced process. Its histological appearance may be related to incomplete bacterial degradation and altered host response." The same study documented that "histiocytes predominated, and their cytoplasm contained numerous lipid droplets with and without limiting membranes and phagolysosomes."

Categories mapped to the chain

- **Cellular processes:** chronic granulomatous inflammation, macrophage activation (GO:0042116), phagocytosis (GO:0006909), foam cell differentiation (GO:0090077), fibrosis.
- **Immune involvement:** an **incomplete/aberrant immune response** is central—chronic inflammation rather than autoimmunity or classical immunodeficiency PMID: 34522121; PMID: 39185472.
- **Protein/biochemical:** macrophage lipid handling defect leading to intracellular lipid droplet accumulation (foam-cell formation); *Proteus* urease-driven urinary alkalinization and struvite lithogenesis.
- **Tissue-damage mechanisms:** suppuration/abscess, granulomatous destruction, fibrosis, and progressive loss of functional nephrons.
- **Metabolic:** local lipid accumulation within histiocytes; no systemic metabolic defect.

Suggested ontology terms

- **GO (biological process):** inflammatory response (GO:0006954), macrophage activation involved in immune response (GO:0002281), foam cell differentiation (GO:0090077), response to bacterium (GO:0009617).
- **Cell types (CL):** macrophage (CL:0000235), foam cell (CL:0000891), neutrophil (CL:0000775), plasma cell (CL:0000786), lymphocyte (CL:0000542), macrophage giant cell.
- **CHEBI:** cholesterol / lipid (CHEBI:16113), struvite (magnesium ammonium phosphate).

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7. Anatomical Structures Affected

Organ level. - **Primary organ:** kidney (UBERON:0002113), specifically the renal parenchyma and pelvicalyceal system. Involvement is **usually unilateral** and produces an enlarged, non-functioning kidney. Both kidneys are affected at roughly equal frequency (right 45.5%, left 54.5% in one series) PMID: 41250779. - **Secondary / adjacent involvement:** perinephric fat, psoas muscle (UBERON:0001096), colon (renocolic fistula), skin (nephrocutaneous fistula), and rarely the diaphragm/lung (transdiaphragmatic extension with lung abscess) PMID: 36105829; PMID: 34277481; PMID: 37171318. - **Body systems:** urinary/renal system primarily; secondary involvement of the musculoskeletal, gastrointestinal, integumentary, and respiratory systems via local extension; systemic (sepsis in ~13% of a large series) PMID: 37242365.

Tissue and cell level. Renal parenchyma (tubulointerstitium) is destroyed and replaced by sheets of foamy histiocytes with granulomatous inflammation. Key cell populations: **CD68+ lipid-laden macrophages/foam cells** (CL:0000891), neutrophils, lymphocytes, plasma cells, and multinucleated giant cells PMID: 21526966; PMID: 42288880.

Subcellular level. Lipid droplets and **phagolysosomes** within histiocyte cytoplasm are the ultrastructural hallmark; bacteria persist within phagocytic vacuoles PMID: 7076194. Relevant GO cellular components: lysosome (GO:0005764), phagocytic vesicle (GO:0045335), lipid droplet (GO:0005811).

Localization / lateralization. Predominantly **unilateral**; may be **diffuse** (whole kidney) or **focal/segmental** (a portion of the kidney, more common in children). Bilateral or horseshoe-kidney involvement is rare but a strong predictor of adverse outcome PMID: 39254583.

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8. Temporal Development

Onset. Chronic and typically **insidious**; symptoms (flank pain, low-grade fever, recurrent UTI) may be attributed to nephrolithiasis or simple UTI, delaying diagnosis (median diagnostic delay ~365 days) PMID: 36505992; PMID: 36473546. Age of onset is most often adult but ranges from neonate to elderly PMID: 21526966.

Progression. The disease is **progressive** with ongoing parenchymal destruction if untreated. Two anatomic patterns are recognized: **focal/segmental** (earlier, more amenable to kidney-sparing therapy) and **diffuse** (advanced, usually requiring nephrectomy). Advanced disease can become fulminant with sepsis, abscess, and fistula formation.

Duration and course. Without definitive treatment the course is chronic and typically culminates in a non-functioning kidney and/or invasive complications; with nephrectomy the disease is essentially **cured** (it does not recur in the removed kidney).

Critical periods / windows for intervention. Early recognition—prompted by awareness of risk factors (stones, recurrent UTI)—allows nephron-sparing strategies and prevents late-stage complications PMID: 36473546. Once diffuse destruction or invasive complications occur, nephrectomy becomes both necessary and technically more difficult.

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9. Inheritance and Population

Inheritance. None—XGP is **acquired and non-heritable**. There is no inheritance pattern, penetrance, expressivity, anticipation, mosaicism, founder effect, consanguinity role, or carrier frequency to report.

Epidemiology. XGP is **rare**; no reliable population incidence/prevalence rates exist. It is estimated to account for a small percentage of chronic pyelonephritis/nephrectomy specimens. In surgical series of poorly/non-functioning kidneys, XGP represented ~14% of nephrectomy histopathology in one large audit PMID: 39558456.

Demographics. - **Sex:** female predominance—62.5% women in a 365-patient multicenter series PMID: 37242365; 57.4% in a 251-patient pooled analysis PMID: 39254583; 71.9% in a 57-patient series PMID: 36505992. - **Age:** middle-aged adults predominate—mean age 45 ± 14.4 years PMID: 37242365; 36.1 years in the pooled cohort PMID: 39254583; range newborn to elderly. - **Geographic distribution:** worldwide, with substantial case literature from South Asia, the Middle East, and tropical regions where stone disease and delayed presentation are common PMID: 41250779; PMID: 42136792.

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10. Diagnostics

Definitive diagnosis is histopathologic and often requires immunohistochemistry, because XGP overlaps clinically and radiologically with RCC, renal tuberculosis, and pyonephrosis PMID: 41250779.

Laboratory tests. Anemia (78.9%), leukocytosis with left shift, elevated inflammatory markers; urinalysis shows pyuria/leukocyturia (93.2%), hematuria (75.6%), and bacteriuria (40.9%) PMID: 36505992. **Urine culture** most often grows *E. coli* or *Proteus mirabilis*; note that **upper-tract (kidney) cultures are positive far more often (81.9%) than bladder cultures (53.2%)** PMID: 37242365.

Imaging. CT is the mainstay. Classic features: an enlarged non-functioning kidney with a central **staghorn calculus**, hydronephrosis, and multiple low-attenuation areas (dilated calyces/abscess cavities) producing the **"bear paw" sign** PMID: 34522121; PMID: 41323310. CT also stages perinephric/psoas extension and fistulae. Imaging cannot reliably exclude malignancy (RCC/urothelial/squamous carcinoma can mimic the bear-paw appearance) PMID: 42558360.

Biopsy / pathology. Histology reveals **sheets of foamy histiocytes** with multinucleated giant cells, neutrophils, lymphocytes, plasma cells, and fibrosis, replacing parenchyma PMID: 21526966; PMID: 41250779; PMID: 42110084. "The characteristic pathological finding is the presence of foam cells" PMID: 3750578.

Immunohistochemistry. CD68 confirms the macrophage/histiocyte lineage of the foam cells PMID: 42288880. IHC is often **decisive** in distinguishing XGP from RCC: "The misinterpretation of 'foam cells' as 'clear cells' consistent with RCC is the most important diagnostic challenge for pathologists," and "The immunohistochemistry was the decisive factor in confirming the diagnosis of XGPN" PMID: 37113900.

Genetic / omics diagnostics. Not applicable—no genetic test, panel, karyotype, or omics assay is used to diagnose XGP.

Differential diagnosis. Clear-cell/papillary/sarcomatoid RCC, squamous cell carcinoma and urothelial carcinoma of the renal pelvis, renal tuberculosis, pyonephrosis, malakoplakia, megalocytic interstitial nephritis, and leiomyosarcoma PMID: 21526966. Red flags favoring malignancy include culture-negative, antibiotic-refractory disease with an irregular enhancing solid component and vascular involvement PMID: 42558360.

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11. Outcome / Prognosis

Overall prognosis. XGP is **curable with nephrectomy** but carries **high perioperative morbidity**. In a pooled analysis of 251 patients, the adverse composite outcome occurred in 15.5% and any complication in 51.0% PMID: 39254583. A dedicated nephrectomy series reported that "a postoperative complication occurred in 73% (n = 8), with 50% (n = 4) of complications Clavien Dindo grade 3 or higher, including one patient mortality" PMID: 37610600. Mortality overall is low (three deaths among 365 patients in a multicenter series) but real, driven by sepsis and advanced disease PMID: 37242365.

Prognostic factors (adjusted odds ratios, pooled n=251) PMID: 39254583:

Predictor	Outcome	OR (95% CI)	p
Bilateral / horseshoe kidney	Adverse composite	3.86 (1.01–14.73)	0.048
Dialysis required	Adverse composite	8.64 (2.27–32.94)	0.002
Staged nephrostomy → nephrectomy	Adverse composite	4.57 (1.58–13.17)	0.01
Fever	Any complication	3.04 (1.63–5.67)	<0.001
Renal stones	Any complication	2.55 (1.35–4.81)	0.004

Additionally, **fever and perinephric inflammatory changes (PIC) on CT predicted poor outcomes** in an independent series PMID: 36505992.

Morbidity / complications. "The most frequent ones are Psoas abscess, perinephric abscess, and sepsis. Nephrocutaneous and renocolic fistulas are less common" PMID: 36105829. XGP "can lead to fatal complications, including perinephric inflammation, psoas abscess, and cutaneous fistula" PMID: 29480882. Emphysematous pyelonephritis can co-occur PMID: 41323310.

Pediatric / focal disease has a favorable prognosis: "Conservative management with antibiotics, with or without drainage, succeeded in 64% of cases, and overall outcomes were favorable, with stable renal function and no reported mortality" PMID: 41341186.

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12. Treatment

Overarching strategy. Culture-guided **antibiotics** plus **source control**. Definitive treatment is stratified by disease extent and patient age.

Nephrectomy (definitive, adult diffuse disease). "Although some cases in children have been successfully treated with antibiotics alone, nephrectomy remains the most common treatment for XGP in adults" PMID: 34522121. Laparoscopic nephrectomy is increasingly used and feasible/safe in experienced centers PMID: 42136792, though XGP raises operative difficulty (adhesions, perinephric inflammation), longer operative times, longer hospital stay, and higher conversion-to-open and complication rates PMID: 39558456; PMID: 40063351. *Suggested NCIT:* Nephrectomy (C15288); Laparoscopic Nephrectomy.

Kidney-sparing / conservative management (pediatric and focal disease). Antibiotics with or without drainage succeeded in **64%** of pediatric cases with favorable outcomes and no mortality PMID: 41341186. Partial nephrectomy / heminephrectomy is an option for focal disease PMID: 42110084; PMID: 6810046. *Suggested NCIT:* Antibiotic Therapy (C15352); Partial Nephrectomy (C51716).

Interventional / supportive. Percutaneous nephrostomy for drainage/decompression and staged control of sepsis before definitive surgery PMID: 34522121; PMID: 42281681. Note, however, that a staged nephrostomy-then-nephrectomy course is itself associated with worse composite outcomes—likely a marker of more severe disease PMID: 39254583. Supportive care includes sepsis management, correction of anemia, and treatment of comorbidities. *Suggested NCIT:* Percutaneous Nephrostomy.

Pharmacogenomics / advanced therapeutics. Not applicable—no gene, cell, RNA, targeted, or immunotherapy is used. Antibiotic selection is guided by urine culture and local resistance profiles PMID: 33827527.

Treatment algorithm (synthesis).

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Suspected XGP (flank pain + fever + recurrent UTI + staghorn calculus)
        │
        ├─ Resuscitate; culture-guided antibiotics; CT staging
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        ├─ Diffuse disease (adult) ──────────► Nephrectomy (laparoscopic/open)
        │                                      ± preop nephrostomy if septic
        │
        ├─ Focal disease / pediatric ────────► Antibiotics ± drainage
        │                                      (partial nephrectomy if fails)
        │
        └─ Culture-negative, antibiotic-refractory, solid enhancing mass,
           vascular invasion ──────────────────► Suspect malignancy → tissue Dx / surgery
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13. Prevention

Primary prevention. Prevent and treat the upstream causes: effective management of **nephrolithiasis** (stone clearance, metabolic stone prevention), relief of **urinary obstruction**, and prompt, complete treatment of **UTIs**. Improving healthcare access reduces the neglected-stone pathway to advanced XGP PMID: 41159347.

Secondary prevention. Early recognition through awareness of risk factors and early imaging in patients with recurrent UTI plus stones can enable nephron-sparing strategies and prevent late complications PMID: 36473546.

Tertiary prevention. In established disease, timely source control (drainage/nephrectomy) prevents sepsis, abscess, and fistula formation.

Immunization / genetic screening / counseling. Not applicable—no vaccine and no heritable component.

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14. Other Species / Natural Disease

XGP occurs naturally in non-human species. The **first documented canine case** was reported in a 6-year-old Pomeranian with an enlarged right kidney and hypoattenuating areas on CT, treated by radical nephrectomy; histology showed parenchymal destruction, abscess, and "infiltration of CD68 + vacuolated macrophages" PMID: 42288880. "This is the first documented case of XGP in a dog."

- **Taxonomy:** *Homo sapiens* (NCBI:txid9606); *Canis lupus familiaris* (NCBI:txid9615).
- **Comparative pathology:** The canine lesion recapitulates the human disease—parenchymal destruction, abscess formation, and CD68+ foam-cell infiltration—supporting conserved macrophage-driven pathophysiology.
- **Zoonosis:** XGP is **not** transmissible/zoonotic; it is a host response to obstruction+infection, not a communicable disease. The causative bacteria (*E. coli*, *Proteus*) are ubiquitous commensals/uropathogens, not species-jumping agents in this context.

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15. Model Organisms

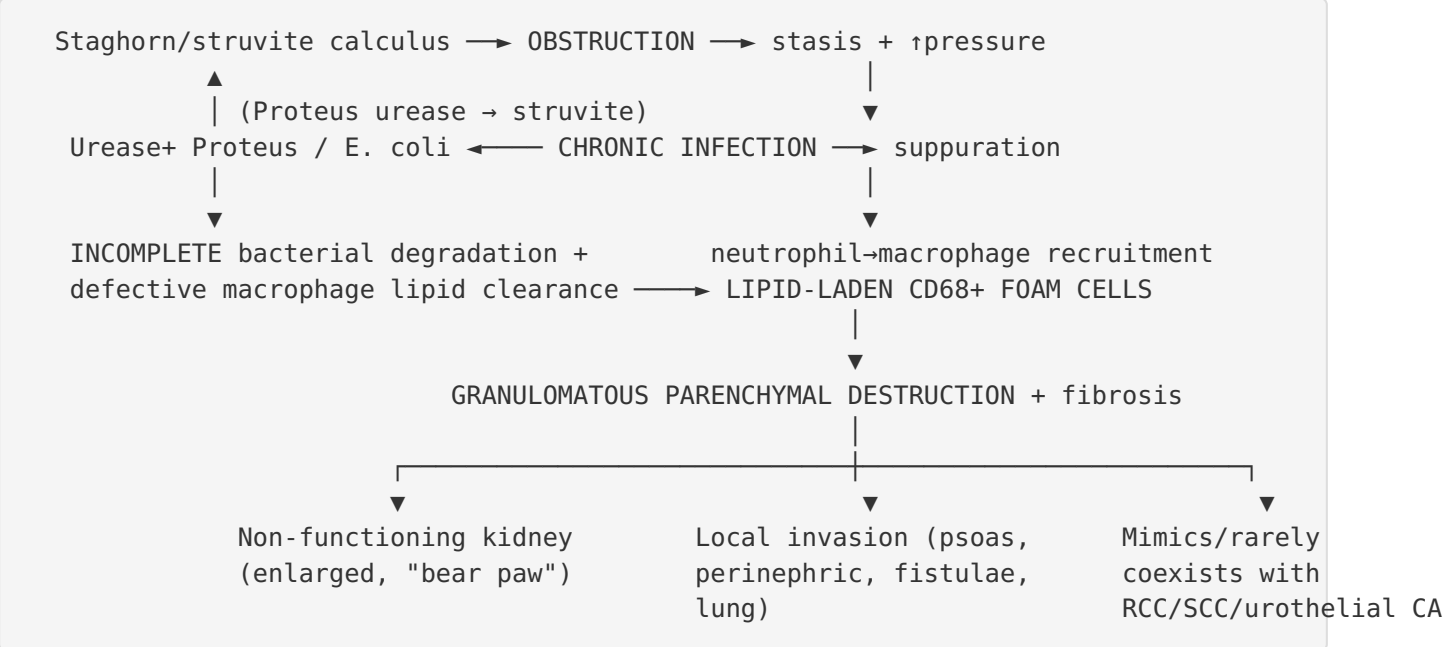
There is **no established, purpose-built animal or in vitro model** of XGP. Knowledge derives from human clinical/pathologic material and the single naturally occurring canine case PMID: 42288880. Historically, experimental partial nephrectomy work was performed in dogs for surgical technique development (not to model XGP biology) PMID: 6810046.

- **Model types:** none validated (no mouse/rat/zebrafish/organoid model of XGP).

- **Phenotype recapitulation:** the naturally occurring canine case reproduces core features (parenchymal destruction, foam cells, CD68 positivity).
- **Limitations / opportunity:** the lack of a tractable model is a significant knowledge gap. A candidate induced model would combine ureteral obstruction with chronic *Proteus mirabilis*/*E. coli* infection in a rodent to test whether foam-cell granulomas form and to dissect the macrophage lipid-handling defect.

Mechanistic Model / Interpretation

XGP is best understood as a **macrophage-centered end-stage response to a self-reinforcing obstruction–infection loop** in a kidney whose immune clearance is incomplete.



Upstream vs downstream. The obstruction–infection loop (with *Proteus* urease closing the loop via struvite lithogenesis) is upstream; the macrophage lipid-handling defect and foam-cell granuloma are the central transducing lesion; parenchymal destruction, non-function, local invasion, and malignancy mimicry are downstream. The **incomplete immune response** is the pivotal host factor that converts ordinary chronic pyelonephritis into the xanthogranulomatous phenotype.

Clinical corollaries that fall out of the model. (1) Upper-tract cultures outperform bladder cultures because the infection is sequestered behind obstruction. (2) Fever and perinephric inflammatory change mark more aggressive host–pathogen interaction and predict worse outcomes. (3) Because the lesion is macrophage/foam-cell rather than epithelial, CD68 IHC resolves the RCC differential. (4) Removing the diseased kidney is curative because it excises the entire self-reinforcing loop.

Evidence Base

PMID	Type	Contribution
21526966	Review	Definition, female predominance, organisms, histologic infiltrate, differential Dx
34522121	Narrative review	Etiologic triad, "great imitator," imaging, adult nephrectomy vs pediatric antibiotics
7076194	Ultrastructural study (n=5)	Bacteria-induced process; incomplete degradation; foam-cell/phagolysosome architecture
37242365	Multicenter series (n=365)	Demographics, CKD comorbidity, kidney vs bladder culture positivity, sepsis, mortality
39254583	Pooled analysis (n=251)	Symptom frequencies; adjusted ORs for adverse outcomes and complications
36505992	Case series (n=57)	Quantitative symptom/lab profile; fever+PIC predict poor outcome; diagnostic delay
33827527	Microbiology series (n=27)	Organism frequencies (Proteus 18.5%, E. coli 14.8%) and resistance
37610600	Surgical series	73% postoperative complications, 50% Clavien-Dindo ≥3, one death
41341186	Pediatric review (n=34)	53% focal; conservative management success 64%, no mortality
42288880	Veterinary case report	First canine XGP; CD68+ foam cells confirm macrophage lineage
37113900	Case report	IHC decisive vs RCC; "foam cells" vs "clear cells" pitfall
42110084	Pediatric surgical report	Histology: parenchymal loss replaced by lipid-laden macrophages + fibrosis
3750578	Review + cases	Foam cells as characteristic pathologic finding
36105829	Case report/review	Complication frequencies; urothelial CA mimicry + renocolic fistula
29480882	Case report	

PMID	Type	Contribution
		Range of local invasive complications incl. cutaneous fistula
33521318, 21119270, 26816904, 21915390	Case reports	Rare coexistence with SCC/collision tumors; urolithiasis–malignancy link
42558360	Case report	Malignancy red flags distinguishing sarcomatoid RCC from XGP
41250779	Case series (n=55)	Clinicopathologic spectrum; histology confirms diagnosis post-nephrectomy

Consistency / conflict assessment. Findings are highly consistent across independent series regarding female predominance, the obstruction+infection+immune triad, *E. coli*/*Proteus* dominance, foam-cell histology, and nephrectomy-based management. The main nuance is treatment: adult diffuse disease → nephrectomy; pediatric/focal disease → conservative therapy often succeeds—these are complementary, not conflicting. No source contradicts the core mechanistic model.

Limitations and Knowledge Gaps

1. **Evidence hierarchy.** The literature is dominated by retrospective series and case reports; there are **no randomized trials** and no prospective natural-history cohorts. Effect estimates (e.g., outcome ORs) derive from pooled published cases and may carry publication bias.
2. **No molecular/omics data.** There are no transcriptomic, proteomic, metabolomic, single-cell, or epigenomic studies specific to XGP. The proposed **macrophage lipid-handling defect** is inferred from ultrastructure, not demonstrated mechanistically (e.g., no identified lipid-transport or efferocytosis lesion).
3. **No genetic dimension.** Host-susceptibility factors (why some obstructed/infected kidneys become xanthogranulomatous and others do not) are unknown; no GWAS or immunogenetic studies exist.
4. **No validated experimental model.** Absence of a tractable animal/organoid model limits mechanistic dissection and therapeutic innovation.

5. **Epidemiology is imprecise.** True incidence/prevalence are unknown; frequencies come from surgical/nephrectomy denominators that overrepresent severe disease.
6. **Malignancy coexistence is under-quantified.** The true rate of concurrent carcinoma is unknown; reliance on case reports risks both over- and under-estimation.

Proposed Follow-up Experiments / Actions

1. **Single-cell / spatial transcriptomics of XGP nephrectomy specimens** vs control chronic pyelonephritis and RCC—define the foam-cell macrophage state (CD68+ subpopulations), test for a specific lipid-handling/efferocytosis signature (e.g., ABCA1, TREM2, APOE, SCARB1), and map the stratified granuloma spatially. *(Resolves the inferred "defective lipid clearance" step.)*
2. **Immunophenotyping of the "incomplete immune response"**—multiplex IHC/flow for macrophage polarization (M1/M2), T-cell subsets, and cytokine milieu to characterize why bacterial clearance is incomplete.
3. **Develop an induced rodent model:** unilateral ureteral obstruction + chronic *Proteus mirabilis* or *E. coli* infection, with readouts for foam-cell granuloma formation; use it to test whether modulating macrophage lipid metabolism prevents xanthogranulomatous transformation.
4. **Prospective multicenter registry** with standardized imaging (CT staging including "bear paw" and perinephric inflammatory change) and outcomes to validate the prognostic ORs (bilateral/horseshoe kidney, dialysis, fever, stones) and refine risk stratification.
5. **Diagnostic-accuracy study of a minimal IHC panel (CD68 ± PAX8/CAIX/RCC markers)** on percutaneous renal mass biopsies to reduce nondiagnostic/false-negative biopsies and pre-operatively separate XGP from RCC, potentially avoiding unnecessary radical surgery.
6. **Systematic review/meta-analysis of laparoscopic vs open nephrectomy** in XGP to quantify conversion rates, complications, and predictors, informing surgical planning.
7. **Antimicrobial stewardship substudy** correlating upper-tract vs bladder culture results and resistance profiles with outcomes, to optimize perioperative antibiotic selection.

Report compiled from 11 confirmed findings and 42 reviewed papers. Evidence source types: predominantly human clinical (multicenter/pooled series, case series, case reports), one veterinary (naturally occurring canine case), and one ultrastructural/in-situ pathology study; no computational or purpose-built model-organism data available for XGP.
